Markscheme

Mathematics: analysis and approaches
Practice question bank

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26. (a)

Consider the function $f(x) = \dfrac{4x^2 + 4x - 2}{x(x-1)(x+1)}$.

Write $f(x)$ as the sum of partial fractions in the form $\dfrac{A}{x} + \dfrac{B}{x-1} + \dfrac{C}{x+1}$.

attempt to write $A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$ *(M1)*

$$4x^2 + 4x - 2 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$$ **A1**

substituting $x = 0$: $-2 = -A \Rightarrow A = 2$ **A1**

substituting $x = 1$: $6 = 2B \Rightarrow B = 3$ **A1**

substituting $x = -1$: $-2 = 2C \Rightarrow C = -1$ **A1**

[5 marks]

(b)

Hence find the exact value of $\displaystyle\int_2^3 f(x)\,dx$.

$$\int f(x)\,dx = 2\ln|x| + 3\ln|x-1| - \ln|x+1|$$ *(M1)*

evaluating between $x = 2$ and $x = 3$ gives $\ln(13.5) \approx 2.603$ **A1**

[2 marks]